

JAYANTH DASAMANTHARAO

New Brunswick, NJ — (848)313-5617 — jayanthdasamantharao@gmail.com — [linkedin.com/in/djayanth](https://www.linkedin.com/in/djayanth)

PROFESSIONAL EXPERIENCE

Applied Machine Learning Engineer, EliteUS Software Solutions Aug 2024 - Present | Edison, NJ

- Designed and deployed enterprise-grade Gen AI backend systems, powering real customer workflows using LLMs, embeddings, retrieval-augmented generation, focusing on scalability, reliability, production quality.
- Designed backend GenAI services with privacy-by-design principles, including secure data access patterns, tenant isolation, and minimal data retention for enterprise workflows.
- Designed and implemented scalable RESTful APIs and microservices using Python and FastAPI to serve LLM-powered capabilities, supporting high availability, versioned deployments, and secure access patterns.
- Built end-to-end ML pipelines for data preparation, embedding generation, retrieval, model fine-tuning, evaluation, and monitoring, deployed across AWS and Azure to support production inference at scale.
- Developed document-AI and semantic search systems using hybrid retrieval, reranking, and vector databases, indexing 10,000+ pages and serving 12,500+ users with low-latency, fault-tolerant inference.

Data Science Intern, Harvest Software Solutions June 2023 - Sept 2023 | Jacksonville, Florida

- Engineered scalable ML pipelines and monitoring systems for large advertising datasets, increasing campaign visibility from 15% to 45% while ensuring reliability and performance.
- Built ML services and NLP APIs using XGBoost, H2O AutoML, and FastAPI, improving sentiment classification accuracy by 65% and campaign spend efficiency by 50%.
- Designed SQL/Python data pipelines for feature engineering, dataset validation, and model training, enabling repeatable experimentation and production handoff of ML models.

Application Development Associate, Accenture June 2021 - Aug 2022 | Hyderabad, India

- Developed and optimized Python-based ETL and data processing pipelines using hash maps, sets, indexed lookups to efficiently clean, deduplicate, and transform large datasets, reducing production issues by 50%.
- Implemented SQL workflows and joins with partitioning, indexing, and window functions, enabling efficient aggregation and analysis over millions of records in AWS Athena and Redshift.
- Applied batching, caching, and algorithmic optimizations for data validation and pipeline execution, improving system performance and reliability by 25%.

TECHNICAL SKILLS

Programming:	Python, R, SQL. Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, FastAPI.
Machine Learning :	Model Fine-tuning, evaluation and experimentation, deployment, Feature Engineering.
GenAI & LLMs:	LLM fine-tuning, embeddings, RAG, agent/tool orchestration, multimodal ML systems.
AI Platforms/Agents:	Agentic systems, GenAI systems, memory architectures, tool calling, guardrails, monitoring.
Backend systems:	Distributed microservices, REST APIs, service orchestration, fault tolerance, deployments.
Cloud & MLOps:	PostgreSQL, Data Warehousing, AWS (Bedrock, Athena, Redshift), GCP, Azure, Vectordb.

EDUCATION

Rutgers University, Master of Science in Data Science Sept 2022 - May 2024 | New Brunswick, NJ
Personal Website: <https://jayanthdasamantharao.github.io/>

Andhra University B.Tech in Electrical and Electronics Engineering 2017 - 2021 | Visakhapatnam, India

RESEARCH PROJECTS

Self-Driving Cars (Computer Vision): Built autonomous driving cars with vision pipeline using CNNs, YOLO, Faster R-CNN, achieving 95%+ object detection accuracy.

Enterprise LLM-Powered Document Retrieval & Generation System (Production): Built LLM-powered retrieval and generation pipelines using embeddings, caching, multimodal inputs, focusing on response quality, robustness, and production reliability.

Bayesian Classification: Implemented Bayesian logistic regression in PyMC3 to predict diabetes status, incorporating informative priors, achieving 0.85 F1-score.

For additional projects and code samples: <https://github.com/JayanthDasamantharao>